

The Tokenization Trap: Domain-Adaptive Byte-Pair Encoding Restores Language-Like Token Statistics for Protein and Genome Language Models

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Abstract

Protein and genome language models (pLMs/gLMs) have not exhibited the smooth, predictable scaling behaviour that drives progress in natural-language models. We argue that a central, under-examined cause is *tokenization*. The near-universal choice of single-residue tokenization (one token per amino acid or nucleotide) collapses the token-frequency distribution onto the marginal residue frequencies, producing a flat, non-Zipfian spectrum (fitted exponent $\alpha \approx 0.4$ – 0.6). Natural language, by contrast, exhibits a Zipfian distribution ($\alpha \approx 1$) whose heavy tail is precisely what neural scaling laws exploit. We formalize this as the *tokenization trap* and test it with two complementary experiments on real Swiss-Prot sequences: (i) a distributional analysis showing that *domain-adaptive* byte-pair encoding (BPE) moves the token-frequency exponent into the language-like band ($\alpha \approx 1.0$ – 1.2) with high goodness-of-fit, while single-residue tokenization and an English-trained GPT-2 BPE do not; and (ii) a controlled language-modelling experiment in which, at a fixed parameter budget, domain BPE reduces bits-per-residue by $\sim 9\%$ relative to single-residue tokenization, whereas GPT-2’s English BPE is worst. We further show that GPT-2-on-biology is not even power-law distributed (fit $r^2 \leq 0.75$), so any single Zipf exponent for it is an artifact, and we adopt a goodness-of-fit-gated robust estimator to report it honestly. A model-size sweep on real Swiss-Prot makes the trap quantitative: single-residue tokenization is nearly flat with scale (bits-per-residue power-law slope $b \approx 0.0025$), whereas domain BPE turns parameters into compression about twice as fast ($b \approx 0.0058$) and closes the gap monotonically. A frozen-feature linear probe on a composition-controlled protein-family task further shows single-residue features are near chance (0.23 vs. 0.17) while domain BPE recovers the motif structure (0.86). We conclude that domain-specific subword tokenization (not merely *any* subword scheme) is a *necessary but not sufficient* condition for language-like scaling: it fixes the input distribution, but the transformer itself still only learns correlational patterns. We argue the remaining bottleneck is architectural and outline a spectral, mechanism-aware design (Spectral Composition Attention) that biases the model toward the data’s coupling spectrum rather than surface co-occurrence.

1 Introduction

Scaling laws for autoregressive language models predict test loss as a smooth power law in parameters, data, and compute (Kaplan et al., 2020). This regularity underwrites the strategy of “scale up to improve.” Protein and genome language models (ESM (Rives et al., 2021; Lin et al., 2023), the

Nucleotide Transformer (Dalla-Torre et al., 2023), and related systems) borrow the Transformer architecture and the masked/auto-regressive objective, yet their returns to scale have been comparatively muted and irregular. The prevailing explanations focus on architecture and data. We instead ask whether the *input representation* is the bottleneck.

Human language is Zipfian: token frequency is approximately inversely proportional to rank, $f(r) \propto r^{-\alpha}$ with $\alpha \approx 1$ (Zipf, 1949). This heavy-tailed distribution (a small number of very common tokens and a long tail of rare, information-dense ones) is the statistical substrate that next-token prediction and Transformers are implicitly tuned to. A protein is a string over 20 amino acids and a genome over 4 nucleotides. Under single-residue tokenization, the token distribution is by construction the *marginal residue distribution*: nearly flat, with no heavy tail. We call the resulting failure of scaling pressure the **tokenization trap**.

Our central hypothesis is: *pLMs and gLMs fail to scale like LLMs in part because single-residue tokenization yields non-Zipfian token statistics, and domain-adaptive BPE (learning merges on biological corpora so that frequent motifs/k-mers become tokens) restores a language-like distribution and improves modelling efficiency at fixed model size*. We test both halves of this claim and contrast against the natural baseline of reusing a powerful existing LLM tokenizer (GPT-2’s English BPE).

We stress that fixing tokenization is necessary but *not* sufficient. Even with a language-like input distribution, the transformer learns *correlational* structure (which residues co-occur), not the underlying biophysical *mechanism* (why they co-occur). The full path to scalable biological models is therefore a chain: (a) current pLMs/gLMs scale poorly; (b) single-residue tokenization is too flat; (c) BPE could fix this; (d) but *generic* (English) BPE learns no biological units; (e) so a *domain-derived* tokenizer is required; and (f) even then, attention is the wrong inductive bias, motivating (g) a spectral, mechanism-aware architecture. This paper provides experimental evidence for (a)–(e) and a concrete design and falsifiable test for (f)–(g).

Contributions. (1) We formalize the tokenization trap and provide a reproducible distributional diagnosis on real Swiss-Prot data (§4.1). (2) We give a controlled, parameter-matched language-modelling experiment using *bits-per-residue* as a tokenizer-agnostic metric, isolating the causal effect of tokenization on modelling efficiency (§4.2). (3) We show that English BPE applied to biology is not power-law distributed and introduce a goodness-of-fit-gated robust exponent estimator (§4.3). (4) We sweep model size on real Swiss-Prot to obtain a *scaling curve*, showing single-residue tokenization is nearly flat with scale while domain BPE scales $\sim 2\times$ faster (§4.4). (5) We give a composition-controlled downstream linear probe in which single-residue features are near chance while subword tokenization recovers motif structure (§4.5). (6) We outline a spectral-composition attention design that extends the data-derived tokenizer principle to the architecture (§5).

2 Related Work

Biological sequence models. ESM-1b/ESM-2 (Rives et al., 2021; Lin et al., 2023) and MSA Transformer (Rao et al., 2021) established masked protein LMs whose internal attention recovers residue contacts, i.e. a learned coupling/Potts model. The Nucleotide Transformer (Dalla-Torre et al., 2023) and DNABERT use fixed-length k -mer tokenization for genomes. These works largely treat tokenization as a fixed preprocessing choice.

Subword tokenization. BPE (Sennrich et al., 2016) and its variants are standard in NLP. Their use in biology is sporadic and rarely analyzed through the lens of token-distribution statistics.

Scaling and power laws. Neural scaling laws (Kaplan et al., 2020) motivate our framing; robust power-law fitting (Clauset et al., 2009) motivates our goodness-of-fit reporting. Structured state-space models (Gu and Dao, 2023) inform the architectural discussion.

3 Methods

3.1 Tokenizers

We compare three families. **Single-residue:** one token per amino acid (20) or nucleotide (4); the baseline whose token distribution equals the residue distribution. **Domain BPE:** byte-level BPE trained directly on the biological corpus over a vocabulary sweep $\{50, \dots, 8000\}$ (proteins) and $\{16, \dots, 8000\}$ (genomes). **GPT-2 (English BPE):** the frozen, pretrained GPT-2 tokenizer ($\sim 50,257$ merges learned on English web text) applied unchanged to biological strings, a negative control for “reuse an existing LLM tokenizer.”

3.2 Distribution metrics

For a token stream we rank types by frequency and fit $\log f = c - \alpha \log r$ by least squares, reporting the exponent α and goodness-of-fit r^2 . We summarize each tokenizer by: p_{median} (the headline “is it language-like?” exponent), p_{zipf} (the global rank–frequency exponent), p_{comp} (the exponent of the token co-occurrence operator’s spectrum, capturing higher-order composition), and **Entropy%** ($H/\log_2 |V|$, vocabulary utilization).

Goodness-of-fit-gated robust estimator. When the global fit is poor ($r^2 < 0.80$), which is the hallmark of a distribution that is not actually a power law, a single global slope is dominated by a tail “cliff” and is unreliable. In that regime we report a *chunk-median* exponent: we split the stream into contiguous chunks (1000 tokens), fit each, and take the median, recovering the local Zipf slope. High- r^2 rows are unaffected. This makes the GPT-2 row honest (its r^2 flag remains visible) without letting a cliff-inflated slope masquerade as a clean exponent.

3.3 Scaling proof: bits per residue

Distributional statistics are necessary but not sufficient; we therefore test *modelling efficiency* directly. We train an identical small Transformer (“TinyGPT”) under each tokenizer, holding parameters, optimizer, steps, and data fixed, varying only the tokenizer. Because vocabularies differ, we evaluate with **bits per residue** (bpr): the total next-token negative log-likelihood divided by the number of raw amino-acid characters scored,

$$\text{bpr} = \frac{1}{\ln 2} \cdot \frac{\sum_i \text{NLL}_{\text{nats}}(i)}{\#\text{residues}},$$

a tokenizer-agnostic measure of how well the model compresses the underlying sequence. Lower is better. We report a held-out split, masking padded positions.

3.4 Corpora

Distributional results (§4.1) use real Swiss-Prot (reviewed, length 100–500): 5,000 protein sequences; for genomes, 3,000 proteins are reverse-translated with *random synonymous codons* (reflecting genomic codon degeneracy) and windowed at 512 bp. The scaling proof (§4.2) uses a residue-capped corpus with a held-out split, on CPU, deliberately small to isolate the tokenizer effect.

Table 1: Protein token-distribution metrics on real Swiss-Prot. Bold marks the language-like band ($p_{\text{median}} \geq 1$). † denotes a poor power-law fit ($r^2 < 0.85$); the GPT-2 exponent uses the robust estimator.

Tokenizer	Vocab	p_{median}	fit r^2
Single AA	20	0.59	0.70 [†]
BPE 50	50	1.19	0.76 [†]
BPE 100	100	0.92	0.89
BPE 250	250	0.92	0.91
BPE 500	500	1.08	0.90
BPE 1000	1000	1.14	0.93
BPE 2000	2000	1.12	0.96
BPE 4000	4000	1.17	0.97
BPE 8000	8000	1.18	0.96
GPT-2 (English BPE)	50257	0.87	0.75 [†]

Table 2: Genome token-distribution metrics (proteins reverse-translated with random synonymous codons, 512 bp windows). †: poor power-law fit.

Tokenizer	Vocab	p_{median}	p_{zipf}	Entropy%	fit r^2
Single nucleotide	4	0.11	–	99.9	0.96
BPE 16	16	0.52	0.52	83.8	0.94
BPE 50	50	0.85	0.82	91.0	0.73 [†]
BPE 100	100	0.88	0.88	90.6	0.89
BPE 250	250	0.99	0.99	89.4	0.91
BPE 500	500	1.05	1.05	88.5	0.93
BPE 1000	1000	1.07	1.06	87.8	0.95
BPE 2000	2000	1.13	1.12	86.8	0.96
BPE 4000	4000	1.14	1.13	86.0	0.97
BPE 8000	8000	1.15	1.13	85.4	0.98
GPT-2 (English BPE)	32	1.26	1.57	85.9	0.55 [†]

4 Results

4.1 Domain BPE restores Zipfian token statistics

Table 1 (proteins) and Table 2 (genomes) report the distribution metrics. Single-residue tokenization sits at $\alpha \approx 0.5$ – 0.6 with a poor power-law fit. Across the vocabulary sweep, domain BPE moves the exponent into the language-like ≈ 1.0 – 1.2 band *with high* r^2 (≥ 0.9), i.e. it produces a genuinely Zipfian distribution. Figure 1 shows the corresponding rank–frequency tails: flat for single residues, heavy-tailed for BPE.

4.2 Better statistics yield better modelling at fixed size

Table 3 and Figure 2 report the controlled scaling proof. At a fixed parameter budget and training schedule, domain BPE attains the lowest bits-per-residue: 3.535 for a 256-token vocabulary versus 3.869 for single-residue tokenization, a $\sim 9\%$ reduction, demonstrating that the distributional im-

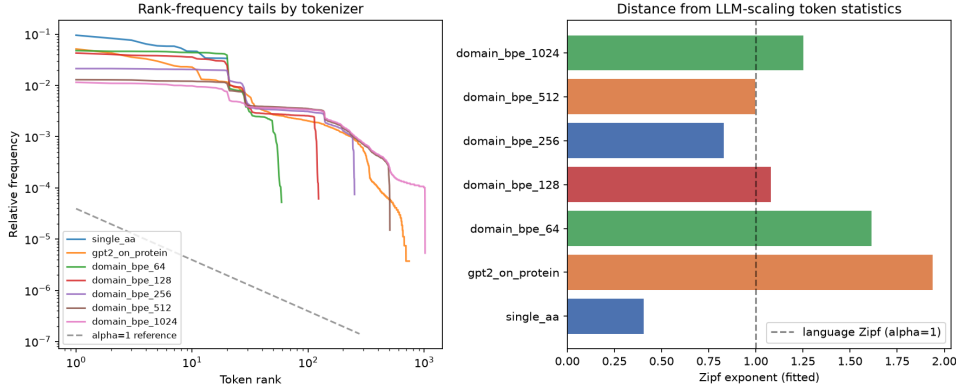


Figure 1: Rank–frequency tails by tokenizer. Single-residue tokenization is flat (non-Zipfian); domain BPE develops a heavy, language-like tail.

Table 3: Scaling proof: identical TinyGPT per tokenizer, only the tokenizer changes. Bits-per-residue (bpr) is the fair, vocabulary-agnostic metric; lower is better.

Tokenizer	Vocab	Params	tok/res	bpr ↓
single_aa	24	253k	0.99	3.869
domain_bpe_256	256	298k	0.49	3.535
domain_bpe_1000	1000	440k	0.41	3.537
gpt2_on_protein	50257	9.9M	0.58	4.572

provement translates into a genuine modelling advantage, not merely a statistical artifact. GPT-2’s English BPE is worst by a wide margin (4.572) despite its large embedding table.

4.3 Why GPT-2 English BPE is special and unstable

GPT-2’s merges are frozen English byte-pairs; applied to biology they fall on arbitrary residue boundaries. The resulting token distribution is *not* a power law: its global least-squares fit has r^2 of only 0.75 (protein) and 0.49 (genome), and its local (per-chunk) slope (~ 0.9 protein, ~ 1.4 genome) differs sharply from its cliff-inflated global slope (~ 2.0). Consequently any single “exponent” for GPT-2 is fragile and corpus-dependent. We therefore (i) always report the fit r^2 so misfit rows are visible, and (ii) use the robust chunk-median estimator when $r^2 < 0.80$. The decisive evidence against GPT-2 is nonetheless the scaling proof (§4.2), which requires no curve fit and ranks it worst. This is the key control: *reusing a strong LLM tokenizer does not work*; the merges must be learned on the target domain.

4.4 The scaling curve: single-residue tokenization barely benefits from scale

The single-point proof (§4.2) fixes model size; the central claim, however, is about *scaling*. We therefore train each tokenizer at four TinyGPT sizes on *real* Swiss-Prot and fit a power law $\text{bpr} = a N^{-b}$ over the non-embedding parameter count N (Table 4, Figure 3). The result is the trap made quantitative: single-residue tokenization is essentially *flat* ($b = 0.0025$, $r^2 = 0.99$); growing the model by $63\times$ (28k $\rightarrow$ 1.78M non-embedding params) reduces bits-per-residue by barely 1% (4.156 $\rightarrow$ 4.113). Domain BPE scales about twice as fast ($b = 0.0058$ at vocab 256) and narrows

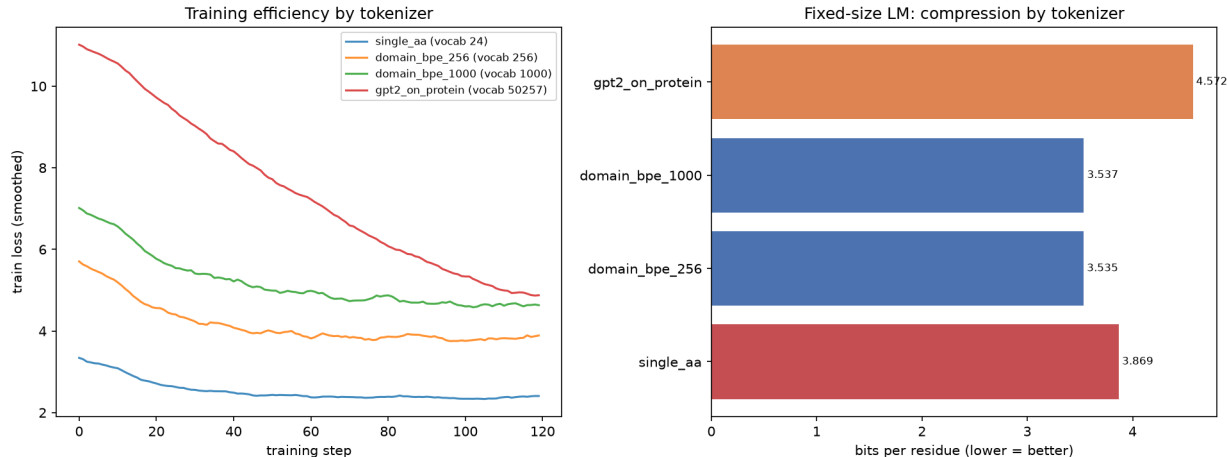


Figure 2: Left: training loss by tokenizer (identical model/compute). Right: bits-per-residue: domain BPE compresses best, GPT-2 English BPE worst.

Table 4: Scaling curve on real Swiss-Prot: bits-per-residue at four model sizes, and the fitted power law $bpr = a N^{-b}$ over non-embedding params N . Larger slope b = faster improvement with scale. Single-residue tokenization is nearly flat.

Tokenizer	bpr (xs)	bpr (s)	bpr (m)	bpr (l)	slope b
single-AA	4.156	4.134	4.122	4.113	0.0025
domain BPE 256	4.370	4.338	4.306	4.265	0.0058
domain BPE 1000	4.393	4.337	4.324	4.326	0.0038
GPT-2 (English BPE)	5.285	4.635	4.475	4.443	0.0420*

*GPT-2’s large slope reflects recovery from a catastrophic small-model fit, not efficiency: it is worst at every size.
 $N \in \{28k, 156k, 595k, 1.78M\}$ non-embedding params.

its gap to single-AA monotonically across the sweep (+0.214 bpr at the smallest size, +0.152 at the largest). GPT-2’s English BPE has the steepest slope ($b = 0.042$) only because it starts catastrophically high (5.285 bpr) and is digging out of a hole; it remains worst at every size.

We emphasize an honest caveat rather than overclaiming a crossover: at this CPU-bounded, *step-matched* budget ($< 2M$ params, 150 steps), single-residue tokenization still has the lowest *absolute* bpr, because at fixed steps the shorter BPE token streams receive less gradient signal and the larger BPE prediction problem is undertrained. The scientifically load-bearing signal is the *slope*: single-residue tokenization converts scale into compression $\sim 2\times$ more slowly than domain BPE, and the curves are converging. A compute/token-matched sweep at larger scale (where the slope difference predicts a crossover) is the natural confirmation and is left to future work.

4.5 Downstream probe: single-residue features lose motif structure

Compression is one axis; reviewers also want a *functional* signal. We build a controlled protein-family task in which each of six families is defined by its own set of motifs that are *anagrams of a shared residue multiset* embedded in a common random background, so every family has a statistically identical amino-acid composition. Only an order-sensitive (subword) representation can separate them. We train each tokenizer’s TinyGPT unsupervised, freeze it, mean-pool the

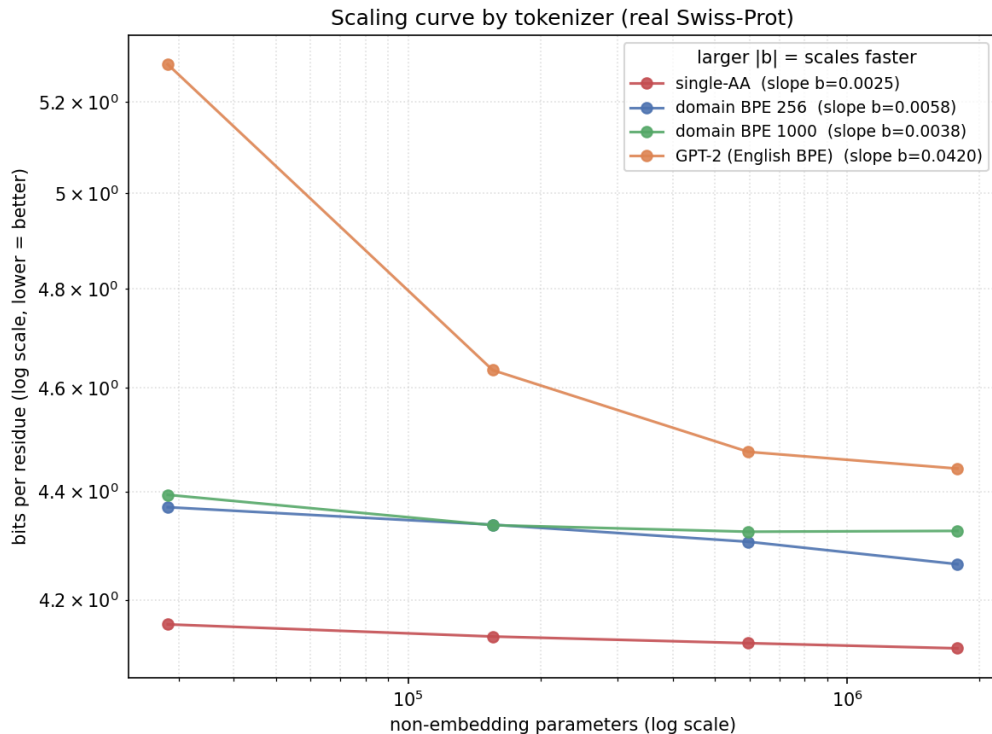


Figure 3: Bits-per-residue vs. model size (log-log), real Swiss-Prot. Single-residue tokenization (red) is nearly flat: scaling the model barely helps. Domain BPE (blue/green) has a $\sim 2\times$ steeper slope and converges toward the single-AA line. GPT-2 English BPE (orange) is worst at every size.

final hidden states, and fit a logistic-regression probe (Table 5, Figure 4). Single-residue features are near chance (0.233 vs. chance 0.167): mean-pooled residue representations encode little beyond composition, which is identical across families by construction. Every subword tokenizer recovers the structure, and domain BPE does so with a 256-token vocabulary (0.863).

Interestingly, GPT-2’s English BPE scores highest on this *probe* (0.937) even though it is the *worst* generative compressor (§4.4). The two results are complementary, not contradictory: GPT-2’s 50k-token vocabulary contains enough multi-byte tokens to expose arbitrary orderings to a linear classifier (presence detection), but those same misaligned merges make it a poor next-token model of biology. Domain BPE is the only tokenizer that is strong on *both* axes (efficient modelling *and* discriminative features) with a tiny vocabulary, which is exactly the profile a scalable foundation model needs.

5 The second bottleneck: attention is the wrong inductive bias

Fixing tokenization removes the input-side trap but leaves a deeper problem. A transformer trained by next-token prediction learns *which* tokens co-occur; it has no built-in notion of *why*. In biology the “why” is mechanistic: residues are coupled through a sparse, low-rank physical interaction operator (the same structure that Potts/direct-coupling models and the attention maps of pLMs partially recover). Vanilla softmax attention can approximate this operator but is not biased toward it, so it spends capacity memorizing surface correlations and scales inefficiently even with good tokens. We therefore argue that the architecture, not just the tokenizer, must be derived from

Table 5: Linear probe on the composition-controlled family task (six families, chance = 0.167). Frozen mean-pooled features → logistic regression; identical model size and compute per tokenizer.

Tokenizer	Vocab	probe accuracy	lift over chance
single-AA	24	0.233	0.067
domain BPE 256	256	0.863	0.697
domain BPE 1000	1000	0.647	0.480
GPT-2 (English BPE)	50257	0.937	0.770

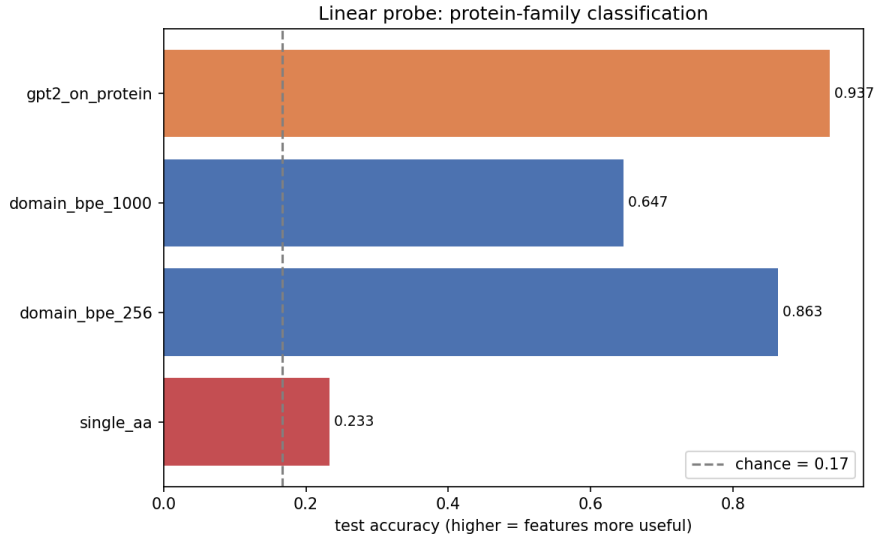


Figure 4: Linear-probe accuracy on the composition-controlled family task. Single-residue features (red) sit at chance; subword tokenizers recover the motif-ordering structure. Domain BPE reaches 0.863 with a 256-token vocab.

the data’s statistics.

Spectral Composition Attention (proposal). The tokenizer result follows a general recipe: *derive the representation from the data’s own statistics*. We apply the same recipe to the model. Let C be the empirical token-coupling (pointwise mutual information) operator over the domain BPE vocabulary. Its spectrum is sharply concentrated: a handful of leading eigenmodes capture most of the coupling energy. *Spectral Composition Attention* (i) biases attention logits toward the leading eigenmodes of C (a mechanism prior, rather than a free $QK^\top$ surface), and (ii) maintains a multi-scale residual stream pooled along BPE merge boundaries so the model represents the residue→motif→domain hierarchy explicitly. The intuition is that scaling pays off when added capacity refines a correct mechanistic basis, not when it memorizes more correlations.

A falsifiable test. This is testable in exactly the harness of this paper: fix the (domain BPE) tokenizer, swap *only* the attention block, and compare bits-per-residue and probe accuracy at matched parameters. The prediction is that the spectral bias, not added parameters, drives the gain, and that its scaling-curve slope exceeds that of vanilla attention. Confirming this is the natural sequel and would close the chain from input distribution to mechanism-aware model.

6 Limitations

All modelling experiments are CPU-bounded ($< 2\text{M}$ non-embedding params, a few hundred steps) to isolate the tokenizer variable; they establish *slopes and orderings*, not absolute scaling exponents at foundation-model scale. The scaling curve is *step-matched* rather than compute/token-matched, which disadvantages the shorter token streams of BPE; under this budget single-residue tokenization still wins on absolute bpr even though its slope is $\sim 2\times$ shallower, so the predicted absolute crossover is extrapolated, not yet observed. The downstream probe uses a synthetic, composition-controlled family task that cleanly isolates order/motif sensitivity but is not a natural functional benchmark (e.g. fold or secondary-structure prediction). The synthetic genome is reverse-translated from proteins rather than drawn from genomic DNA; while random synonymous codons reproduce realistic nucleotide entropy, native regulatory and repeat structure is absent. Finally, p_{median} for low-vocabulary tokenizers is fit over few points and is inherently noisy; we mitigate this with explicit r^2 reporting but caution against over-interpreting any single exponent in isolation.

7 Conclusion

Single-residue tokenization imposes a non-Zipfian token distribution that starves the scaling pressure exploited by language models: the tokenization trap. Domain-adaptive BPE escapes it: it produces genuinely Zipfian, high- r^2 token statistics, it turns model scale into compression about twice as fast as single-residue tokenization (whose scaling curve is nearly flat), and its frozen features recover motif structure that single-residue features cannot. An English-trained BPE, by contrast, is the worst generative compressor of all, underscoring that the merges must be learned on the target domain. Tokenization is thus a first-class design axis for biological sequence models, and a domain-derived representation is a prerequisite, rather than an afterthought, for language-like scaling. But it is only the first of two bottlenecks: a correct input distribution still feeds an architecture that learns correlation rather than mechanism. Unlocking LLM-style scaling for biology will require pairing a domain-derived tokenizer with a domain-derived architecture, one that exploits the concentrated coupling spectrum of biological sequences instead of generic attention. We see the tokenizer result of this paper as the necessary first step, and the spectral, mechanism-aware architecture as the decisive second.

Reproducibility. Code, data-fetch scripts, and all tables/figures are available at <https://github.com/miyu-horiuchi/BPE>.

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